ACETYLCYSTEINE

Paracetamol antidote

Anaphylactoid reactions

CHEMICAL NAME (2R)-2-acetamido-3-sulfanylpropanoic acid (N-acetyl-L-cysteine, NAC)
CLASS Antidote (glutathione precursor); mucolytic
TRADE NAMES Parvolex · Acetadote · Fluimucil · NAC (generic)

THERAPEUTIC USES / INDICATIONS

- Paracetamol overdose — prevents hepatotoxicity (most effective within 8 h, but give at any time if indicated).
- Mucolytic in COPD/bronchiectasis/cystic fibrosis (oral/nebulised).
- Prevention of contrast nephropathy (evidence weak); acute liver failure (non-paracetamol, some protocols).

ADMINISTRATION

- **IV** infusion in 5% glucose (or saline) by protocol — e.g. SNAP 12-hour regimen or 21-hour three-bag regimen.
- Use weight (max 110 kg) for calculations; double-check volumes.
- **PO** effervescent tablets/sachets (mucolytic); **nebulised** with bronchodilator (bronchospasm risk).

DOSAGES

- **Paracetamol overdose (21-h regimen):** 150 mg/kg over 1 h, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h.
- **SNAP 12-h regimen:** 100 mg/kg over 2 h, then 200 mg/kg over 10 h.
- Treatment decision from timed paracetamol level (≥4 h) vs nomogram (treatment line 100 mg/L at 4 h in UK), or immediately if staggered/unknown time or >8 h.
- **Mucolytic:** 600 mg once daily PO (or 200 mg three times daily).

PHARMACOKINETICS

- **Onset:** IV immediate.
- Deacetylated to cysteine (glutathione precursor) in liver.
- Excreted in urine. **t½:** 2–6 h.

PHARMACODYNAMICS

- Replenishes hepatic **glutathione** → conjugates and detoxifies toxic paracetamol metabolite NAPQI; antioxidant.
- Breaks disulfide bonds in mucus → ↓ viscosity (mucolytic).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting.
- Flushing, rash, itching (common).
- Unpleasant smell (sulfur).

ADVERSE EFFECTS (SERIOUS)

- Non-IgE anaphylactoid reactions (rash, angioedema, bronchospasm, hypotension); mostly during the first (loading) bag; usually managed by stopping temporarily, antihistamine, then restarting at slower rate.
- Bronchospasm with nebulised use.
- Fluid overload, hyponatraemia in children (volume).

CONTRAINDICATIONS & CAUTIONS

- No absolute contraindication in paracetamol overdose (benefit > risk).
- **Caution:** asthma (nebulised/IV reactions).

KEY INTERACTIONS

- **Activated charcoal:** may reduce oral NAC (not relevant for IV).
- **Nitroglycerin:** ↑ hypotension/headache (reported).

NURSING MANAGEMENT

- Record time of ingestion; take paracetamol level at ≥4 h, plus LFTs, INR, U&E, glucose, VBG.
- Do not delay NAC if >8 h or staggered overdose.
- Monitor vital signs, rash, wheeze, especially in first hour; stop and treat reactions.
- Repeat bloods at end of infusion (ALT, INR, paracetamol) to decide on continuation.
- Psychosocial/mental health assessment.

PATIENT EDUCATION

- Explain why treatment continues for hours even if you feel well.
- Report itching, rash, swelling or breathlessness during the drip.
- Seek help and support; paracetamol overdose can cause delayed liver failure.

KEY POINTS

- Glutathione precursor; antidote to paracetamol (NAPQI).
- 21-h (150/50/100) or 12-h SNAP regimens.
- Use nomogram; treat urgently if >8 h or staggered.
- Anaphylactoid reactions — pause, treat, restart.

SOURCES CHECKED **[992]** Acetylcysteine 200mg/ml Injection (UK SmPC) **[993]** Acetylcysteine, IV (US label) **[994]** Acetylcysteine, oral, inhaled, (inhalation) (US label)
Full references p. 432–457